1. Introduction

Dissolved Oxygen (DO) is the amount of free oxygen gas dissolved in a given body of water. DO is a useful measure of water quality, as it supports aquatic life. With low levels of DO, entire aquatic ecosystems can be damaged, leading to local extinction of key species. Beyond this, low levels of DO can foster bacteria growth and cause hydrogen sulfide emissions, creating foul-smelling air pollution. Furthermore, these bacteria feed on oxygen, continuing the cycle.

The initial cause of low DO can be attributed to a variety of factors. First, agricultural runoff (fertilizer) can facilitate the growth of algae (algal bloom), which lead to dead plant material and bacteria. As mentioned before, the bacteria readily consume oxygen, decreasing DO. Another possible cause is climate change, as oxygen precipitates out of water more easily at higher temperatures (Henry's law). Furthermore, salinity can affect DO, as saltwater holds less oxygen than fresh water. Lastly, decreased DO has been associated with a decrease in pH (increase in acidity), though no chemical connection exists between the two.

While it may seem straightforward to calculate DO based on factors such as salinity, temperature, and pH, their relationships have secondary factors that make it nearly impossible to predict. For example, algal bloom can decrease dissolved CO_2 (thus decreasing acidity and increasing pH). This means that agricultural runoff can cause a short-term increase in DO, before decreasing in the long-term. To account for this complexity, we will use machine learning to calculate the contribution of different factors to DO.

2. Data

We will use the Water Quality Prediction dataset from the UCI Machine Learning Repository, and we will use the factors of pH, specific conductivity (a measure of salinity), temperature, and month of the year.

In the located .zip file, there is a "train" and a "test" set of data. For the sake of this project, I have combined them both and then resampled a train and test set.

	date	location id	pH median	SpC max	pH max	pH min	SpC min	SpC mean	DO max	DO mean	DO min	T mean	T min	T max
0	2016-01-28	2198840	0.648148	0.001131	0.884615	0.001120	0.001113	0.677632	0.841463	0.765152	0.787402	0.293750	0.298077	0.276163
1	2016-01-28	2198920	0.648148	0.001170	0.871795	0.001159	0.001152	0.703947	0.829268	0.772727	0.795276	0.293750	0.301282	0.276163
2	2016-01-28	2198950	0.648148	0.001326	0.884615	0.001198	0.001250	0.677632	0.853659	0.750000	0.755906	0.300000	0.298077	0.287791
3	2016-01-28	2203603	0.638889	0.014094	0.858974	0.001238	0.003926	0.697368	0.829268	0.772727	0.771654	0.296875	0.294872	0.279070
4	2016-01-28	2203655	0.648148	0.088109	0.858974	0.010766	0.029297	0.684211	0.853659	0.765152	0.755906	0.296875	0.291667	0.281977

Figure 1: First 5 entries of the UCI Water Quality Prediction dataset

The combined dataset includes the daily max, min, and median/mean of the variables mentioned earlier (pH, specific conductivity, and temperature), as well as the time and location of the data taken. The quantitative data are already normalized to a value between 0 and 1. For example, pH is typically a value from 0-14, but for this instance, the lowest individual pH datapoint is 0, while the highest is 1, though the median/means are more resistant.

2.1 Proving Location Independence

In order to continue, we must prove that the water quality variables are independent of location. Otherwise, we will have to analyze each location separately with different models. To do this, we can use unsupervised learning (k-means) to cluster the data in 3-dimensional space with DO, SpC, and pH as our three variables. Temperature is left out of this clustering, as it is expected to be consistent across location, and it is impossible to visualize a 4-D k-means clustering. We will choose 37 clusters, then fit them to see if any cluster solely contains points from one location. This would indicate that the location is an outlier and should be removed.

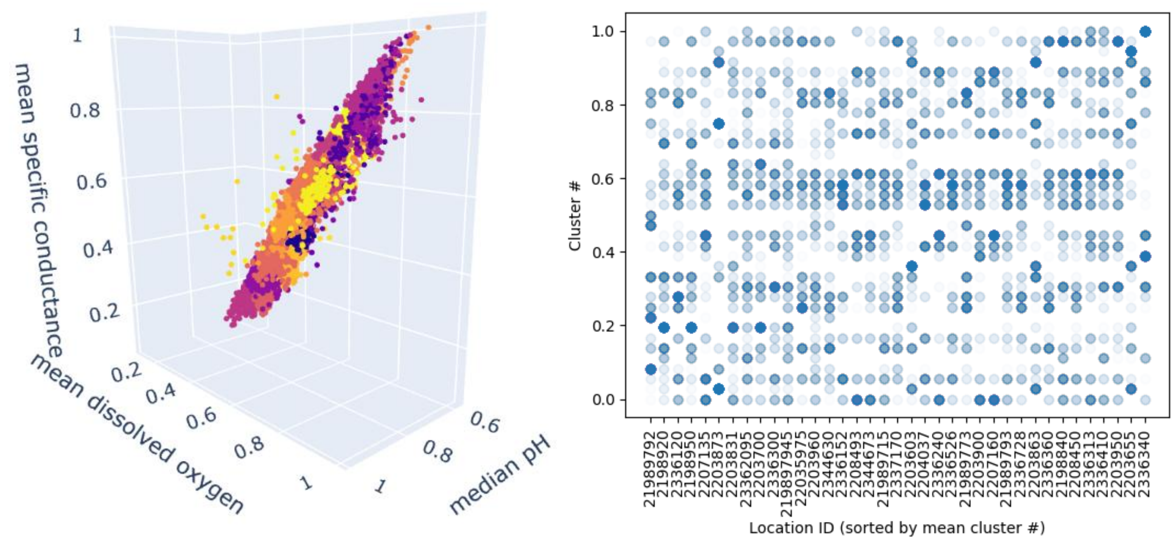

Figure 2. a) 3-D k-means clustering, visualized. Each color represents a different cluster. b) The location ID of each individual point is plotted against its normalized cluster #. The x-axis ticks are then resorted such that the lowest average cluster number is on the left.

From Figure 2a, there are a select few outliers that lay outside the main group of points, indicating a possibility that there is an outlier in the water quality locations. However, Figure 2b appears randomly scattered, showing that there is no correlation between location and the combination of the three factors. Furthermore, when Figure 2b is viewed as columns, there is no location ID with points solely in one cluster (indicating no clear outliers), thus the data can be treated as whole, and no locations need to be removed.

3. Methods

We will use linear regression, ridge regression (α= 10), forest regression, and neural network (MLP) to calculate the associations of pH, specific conductivity (SpC), temperature, and month. For the first three variables, the minimum, maximum, and mean/median value from each day will be added to the model, with the mean DO acting as the target variable. We discard the min and max DO columns, as the goal of the project is to find correlation with non-DO variables.

3.1 Ordinary Least-Squares Regression

The simplest method, thought of as a "baseline" for the performance of others. The cost function is simply the sum of the squared residuals, which is not resistant to outliers. For simplicity, this was taken as ridge regression with $\alpha = 0$.

3.2 Ridge Regression

While similar to least-squares regression, this method accounts for outliers by adding a penalty for large coefficients. This also attempts to decrease the possibility of overfitting. This is controlled by the parameter α, where higher means more penalty. In our case, we choose α=10, a higher value than normal, to see the effect of overfitting.

3.3 Decision Tree Regression

Instead of determining the coefficients for a linear combination of factors, a decision tree chooses a quality with the most importance to sort factors, then makes more decisions to eventually get to a final predicted value. Instead of having coefficient values, there are importance values, which show how often a value is involved in a particular decision. However, this can lead to overfitting and is not resistant to outliers.

3.4 Random Forest Regression

With a fitting name, forest regression involves using many decision trees to bootstrap and then creates a "final" decision tree to avoid overfitting and nonresistance to outliers. This method is more robust than decision tree regression, though it is much more costly.

3.5 Linear Support Vector Regression (LSVR)

Support vector regression is a type of support vector machine (SVM) that attempts to find a "tube" in a higher dimensional space that separates out points, giving penalty to those points outside of the tube. In this case, the tube is linear, meaning that it is possible for outliers to be included.

3.6 Non-linear Support Vector Regression (NLSVR)

This method is very similar to LSVR, however the space that separates points can form any curve, allowing for a more effective separation of points (less resistant to outliers, and better fit for non-linear data). However, due to more hyperparameters and possible equations, this method is more computationally costly.

3.7 Neural Network/Multilayer Perceptron (MLP)

This method relies on hidden layers that improve a guess over time, "learning" the features associated with the problem it was given. While this method is more equipped for "learning" patterns based on large amounts of data (and should not be equipped to solve this problem), we are implementing this anyway to see if this works.

4. Results and Discussion

Method	SpC max	SpC mean	SpC min	pH max	pH median	pH min	T max	T mean	T min	Month
Least-squares	-0.2338	0.9119	0.0554	0.2643	-0.4244	0.1227	-0.102	-0.4392	0.3758	0.0058
Ridge (α=10)	-0.2303	0.7425	-0.0033	0.1075	-0.0883	0.1455	-0.150	-0.0945	0.0056	0.0087
Tree*	0.0044	0.9170	0.0133	0.0018	0.0031	0.0028	0.0160	0.0119	0.0197	0.0010
Forest*	0.0068	0.9153	0.0112	0.0018	0.0028	0.0030	0.0142	0.0265	0.0080	0.0010
LSVR	-0.1915	0.9668	0.0098	0.2771	-0.4135	0.1271	-0.153	-0.2960	0.3172	0.0045
NLSVR*	0.0393	1.3282	0.1173	0.0040	0.0087	0.0322	0.1696	0.3127	0.1705	0.0068
MLP*	0.0653	0.9631	0.0263	0.0007	0.0012	0.0083	0.1628	0.0863	0.0699	0.0034

Table 1: Model coefficients for each variable (for the linear methods). For methods with an asterisk, this is instead the feature importance, which should be treated as an absolute value.

Method	r^2	RMSE	Success Rate
Least-squares	0.9600	0.0294	74.4%
Ridge (α=10)	0.9543	0.0308	69.4%
Decision Tree	0.9739	0.0234	82.8%
Forest	0.9872	0.0165	89.3%
Linear SVR	0.9582	0.0296	75.9%
SVR	0.9784	0.0217	83.0%
MLP	0.9705	0.0252	78.3%

Table 2: Success values associated with different machine-learning methods. Success rate is defined as the percentage of the predicted points that are correctly predicted to within 0.05 of test relative DO values.

5. Conclusion